K1357 — Casey's boundary-refraction reframe of the Weinberg angle (rest-time insight, queued for tomorrow's ballgame). The two $\sin^2\theta$_W values are TWO MEDIA: interior/bulk 3/8 (= Connes, high-energy, complex) vs boundary/measured 3/13 (real, low-energy, where we live as boundary observers). The lens between them is a factor-2 in $\varkappa$ (5/3→10/3) = the complex-to-real doubling = the reality structure J (the same $\sqrt{-1}$ as CP/KO-2). This RE-HOMES the doubling we refuted yesterday: it fails as RUNNING (undershoots to 0.20) but is natural as a BOUNDARY REFRACTION — and it dissolves the undershoot problem because we MEASURE the boundary directly (scale-free, F531), we don't run 3/8 down. A promising LEAD with a sharp falsifiable target; three guards. Cal cold-reads tomorrow.

Keeper (2026-08-10 ~evening, capturing Casey's rest-time reframe. He proposed Snell's-law / changing-media for the two Weinberg values. I corrected one label swap; his structural intuition survives and sharpens. Tiered as a LEAD — attractive, so it gets the hardest scrutiny (Cal #27). Cal cold-reads. Nothing pushed.)

Casey's proposal (corrected labels)

- Casey: "3/13 internal, 3/8 exterior/lab-measured." Correction (factual): lab-measured $\sin^2\theta$_W(M_Z) ≈ 0.231 ≈ 3/13; 3/8 is the high-energy (Connes/GUT) value, never directly measured. So the mapping REVERSES to:
  - 3/8 = interior / bulk / high-energy / COMPLEX (Connes = BST here — both spectral actions give 3/8 in the deep interior).
  - 3/13 = boundary / measured / low-energy / REAL (we are boundary observers — matter as record on the conformal boundary, F531).
- His deeper intuition (kept, it's apt): the two values are the angle in TWO MEDIA; the DIFFERENCE encodes the boundary/lens; homeostasis = the interior is held constant, the boundary adjusts to display it.

Why this is the SAVE of yesterday's refuted doubling

Yesterday (K1356) we killed "doubling → 3/13" — but we killed it as a RUNNING effect (run 3/8 down → undershoots to ~0.20). Casey's reframe re-homes the doubling as a BOUNDARY REFRACTION, where it belongs:
- The clean invariant is in $\varkappa = \mathrm{tr}(Y^2)/\mathrm{tr}(T_3^2)$: interior $\varkappa$ = 5/3 (→3/8), boundary $\varkappa$ = 10/3 (→3/13) — a factor of exactly 2. - That 2 = the complex-to-real doubling (complex interior dim n_C=5 → real observed 2·n_C=10) = the reality structure J (KO-dim 2, $J^2=-1$) = the SAME $\sqrt{-1}$ as CP. - The doubling is NOT an RG-running fact (continuous flow) — it's an algebra fact about crossing complex→real (a one-time step at the interface). So it fails as running and succeeds as refraction. Correct home.

Why it dissolves the undershoot problem

- We do NOT run 3/8 down to the measured value (that's what undershot). We measure the BOUNDARY value directly — and the conformal boundary is scale-free (F531, charge as a pure count on the SO(4,2) conformal boundary), so there is nothing to run.
- Homeostasis, precise: the interior/bulk (3/8) is the rigid geometric constant (D_IV^5 fixed by the five integers, spectral action = Connes); the boundary refracts it through the fixed lens ×2 to the observed 3/13. Interior held constant, boundary adjusts. We (boundary observers) read the boundary value.
- The potential DERIVATION: interior 3/8 is forced (spectral action), the lens ×2 is forced (reality structure J), ⍰ boundary 3/13 = refract(3/8, ×2) is FORCED, not fitted. 3/8 refracted through the reality structure = 3/13.

★ THREE GUARDS (this is pretty → hold hardest; the tests that make it real or kill it)

1. Is the lens FORCED to be exactly ×2 by the complex→real / KO-2 structure — or chosen because it hits 3/13? The 2 must fall out of J / the reality structure BLIND, not be reverse-engineered. This is the whole game. (Target-innocence: don't reason toward 3/13.)
2. Is the boundary 3/13 genuinely SCALE-FREE? Yesterday's open question, unchanged. F531 (conformal boundary, scale-free) says yes; but my #85 corpus note had 3/13 "at the geometric scale, deviations = running" — which would make it NOT scale-free. Confirm, don't assume. If it runs, the whole picture fails.
3. It's NOT literally Snell's law. The conserved/clean quantity is κ (factor 2), not an "n·sinθ" invariant across the interface ($\sqrt{}$(sin^2_bulk/sin^2_bdy) = $\sqrt{}$(13/8) is not clean). So the PICTURE (two media, a boundary lens, index 2) is right; the specific law is a refraction in the *normalization*, not the sine. State it that way — don't oversell "Snell."

- Meta (the honest flag): this makes the Weinberg angle a FOURTH face of the one sign (after reality class, CP, KO-dim). "One sign, another face" has been the weekend's most seductive pattern — so this gets the hardest scrutiny, precisely because it's the most attractive.

Tomorrow's ballgame, sharpened (replaces the K1356 framing with a mechanism)

Not just "pin the scale of 3/13" but: is 3/13 the boundary refraction of the interior 3/8 through a forced ×2 lens? - Elie/Grace: compute the bulk→boundary map for κ BLIND — is the factor exactly 2, and is it FORCED by the complex→real / reality-structure (J) crossing? And confirm the boundary observable is genuinely scale-free (F531). - Lyra: the physical picture — bulk (complex, 3/8, = Connes) → conformal boundary (real, 3/13, measured); the lens = J; homeostasis = rigid interior, refracting boundary; we are boundary observers (matter-as-record, F531). - Cal: cold-read the ×2-forced claim and the scale-free claim; kill it if the 2 is fudged or if 3/13 secretly runs. - Corpus-first (Linear Algebra on D_IV^5): F531 (charge as boundary count, scale-free), #85 (curvature grading 3/13 = N_c/(N_c+2n_C)), the bulk/boundary SO(5,2)⊃SO(4,2) structure, J / KO-2 (the lens). - Win = 3/13 derived as 3/8 refracted through a blind-forced ×2 reality-structure lens, with the boundary confirmed scale-free. Then the Weinberg angle is (honestly) a fourth face of the sign. Anything hand-fit → reported straight.

ADDENDUM (Casey, same evening) — the MECHANISM for the lens: curvature flattening in response to exterior heat

Casey's second intuition gives the lens its dynamics: "the curvature/lens flattening to respond to the exterior heat." This maps to REAL structure, not metaphor: - The Weinberg angle IS a curvature grading (e_5/e_3, #85) — so "curvature flattens" = "the grading changes" = "the Weinberg angle changes." Same object. - "Heat" = the heat-kernel proper time (BST's clock; proper time = energy scale). The curvature term is the a_2 coefficient of the heat expansion. So "curvature responding to heat" = a_2's dependence on heat-kernel time — a concrete object (#17/#94, the induced curvature/β response). - Hot UV interior (curved, 3/8) → cold IR boundary (flat, 3/13): the geometry flattens toward the Minkowski exterior as heat is shed. Consistent with BST's emergent-flat-spacetime-at-low-energy. - ★ It NESTS the two pictures (resolves the A-vs-B tension in guard 2): the FLATTENING is the interior→boundary transition (the lens in motion); the ×2 is its net ENDPOINT; the SCALE-FREE 3/13 is the flat exterior AFTER flattening (the regime we measure in — nothing left to run). Flattening = how; ×2 = where it lands; flatness = the medium we read it in. Homeostasis = interior setpoint (3/8, rigid) held while the boundary flattens to meet the cold exterior, displaying 3/13. - ★ GUARD (added, the honest risk): thermodynamic/homeostatic stories are the MOST seductive — they fit almost anything post-hoc. We now have THREE stacked framings (refraction + curvature-flattening + homeostasis); the discipline is to COLLAPSE to the one forcing computation, not admire the stack. The test (unchanged, now with a handle): does the curvature grading flow from 3/8 (hot interior) to 3/13 (cold boundary) with the endpoint FORCED (not tuned) and SCALE-FREE once flat — computed from the a_2 curvature response BLIND? If the ×2 falls out of the flattening geometry → derivation with a mechanism. If the stopping point is hand-set → a story that fit the known answer; report it straight.

— Keeper, K1357, 2026-08-10. Casey's boundary-refraction reframe (rest-time) + curvature-flattening mechanism addendum. Corrected: 3/8 = interior/bulk/high-E (Connes), 3/13 = boundary/measured/low-E (we're boundary observers). The lens between = factor-2 in $\varkappa$ (5/3→10/3) = complex→real doubling = J ($\sqrt{-1}$, CP/KO-2). RE-HOMES the doubling refuted yesterday: fails as RUNNING (undershoot to 0.20), natural as BOUNDARY REFRACTION; dissolves undershoot because we measure the scale-free boundary (F531) directly, don't run 3/8 down. Homeostasis: rigid interior 3/8, boundary refracts ×2 → 3/13. Potential derivation: 3/8 (forced) refracted through ×2 (forced J) = 3/13. THREE GUARDS: (1) is ×2 FORCED not chosen (the whole game, target-innocent)? (2) is 3/13 genuinely scale-free (F531 yes vs #85 "runs" — confirm)? (3) not literal Snell (invariant is $\varkappa$ not $\sin\theta$). Meta: 4th face of the sign → hardest scrutiny. Tomorrow: compute bulk→boundary $\varkappa$-map blind, ×2-forced?, scale-free?; win = 3/13 derived as refracted 3/8. Cal cold-reads. Nothing pushed.